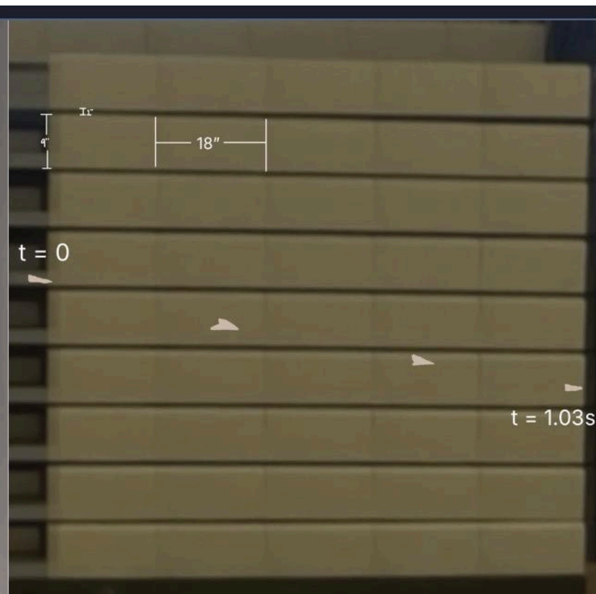
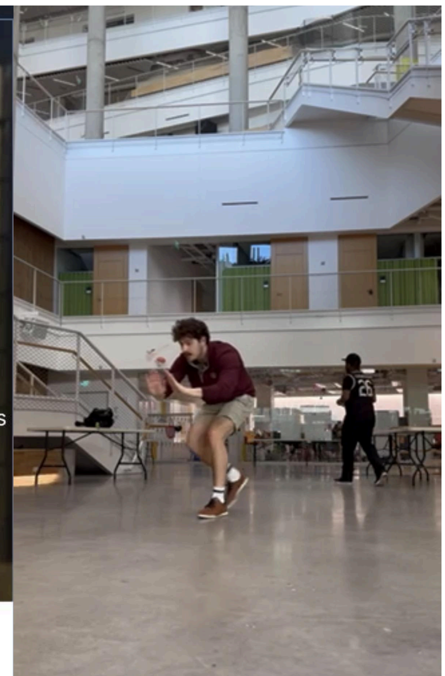
In the Spring semester of my junior year, I petitioned the mechanical engineering department heads to allow me to enroll in ES220: Fluid Dynamics Across Scales, Harvard's graduate-level fluid mechanics course, instead of taking ES123, the standard fluid mechanics class for undergrads. I had many of the physics and engineering prerequisites, and felt that after taking all other core MechE requirements I was ready to take on a greater challenge. Around this time I had also become increasingly interested in aerospace and aeronautical engineering, and I felt that taking ES220 would open the door for more advanced engineering electives at MIT senior year.

The class was 75% project graded, and students formed groups of two to consult with the professor and choose a project to complete, write a paper on, and give a presentation. My group chose the task of studying walkalong gliders, a lightweight gliding airfoil designed to be able to maintain steady lift in a region of upwash generated by the user walking behind and underneath the glider, as shown in the GIF on the main page. This concept was first invented by Tyler McCready, and he can be seen flying his glider on youtube:
https://youtu.be/S6JKwzK37_8?si=QSoGhdN0jZPrLVSI

The overall project included an overview of the equations and theory governing the glider flight, actually building and flying a glider, and testing and iterating those designs. While building a glider that one could fly in a user-generated upwash region was not too difficult, building one that could be flown without a large flat sheet of material to better create upwash and one that could be flown at walking pace proved much more difficult. I tested more than 4 different materials and more than 20 different airfoil shapes, taking inspiration from online sources and from my own intuition. This project stands out and was so enjoyable for me not only because of the sensitivity to geometry and need for much iteration of the airfoil, but because the design choices I made and which eventually lead to a final product, were largely driven by aerodynamic intuition I had honed in class, and observations I made while trying and failing to fly many different designs. The project required physical knowledge and intuition, but also creativity, testing and iteration, and tenacity, and those are the qualities that drive me towards engineering.

The final paper and presentation will give you a better idea of the project as a whole, but here are some photos from along the way:





Quantity	Final light model	Heavier comparison model
Mass m	0.94 g	1.68 g
Projected area S	0.011 m^2	0.011 m^2
Chord Length c	0.0635 m	0.0635 m
Glide slope $\tan\gamma$	0.20	0.25
Glide speed V	2.2 m/s	2.8 m/s
Lift coefficient C_L	0.265	0.558
Drag coefficient C_D	0.0531	0.140
Re	9.4×10^3	1.2×10^4